Assistant Professor
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RESEARCH INTERESTS

- Finite element and discontinuous Galerkin methods
- Scientific machine learning and data-driven approaches
- Symplectic and structure-preserving methods
- Adaptive mesh solver for radiative transfer
- Electromagnetic and elastic/viscoelastic waves
- Computational inverse and ill-posed problems

PROFESSIONAL EXPERIENCE

Syracuse University
- Assistant Professor — Oct 2024 – now

University of Wisconsin-Madison
- Visiting Assistant Professor — Sep 2020 – Sep 2024

University of Minnesota-Twin Cities
- Visiting Doctoral Student — Sep 2019 – June 2020

EDUCATION

University of Delaware
- Ph.D. in Applied Mathematics — 2020
  Advisor: Dr. Francisco-Javier Sayas
  Thesis: Generalized projection-based error analysis of hybridizable discontinuous Galerkin methods

Wuhan University
- M.S. in Computational Mathematics — 2015
- B.S. in Pure Mathematics — 2012

PUBLICATIONS

Submitted

20. S. Du, J. Zhang, and Y. Li. Lagrangian flow matching: a least-action framework for principled path design. arXiv:2605.15419
19. S. Du, T. Le, and S. N. Stechmann. Element learning with hp-adaptivity: machine learning-based acceleration of hp-adaptive spectral element methods, and application to atmospheric radiative transfer.

Peer-reviewed

18. B. Cockburn, S. Du, and M. A. Sánchez. A spectral analysis of hp-hybridized mixed and HDG methods for parametric second-order elliptic problems. SIAM J. Numer. Anal. (2026), to appear.

17. J. L. Torchinsky, S. Du and S. N. Stechmann. Angular-spatial hp-adaptivity for radiative transfer with discontinuous Galerkin spectral element methods. J. Quant. Spectrosc. Radiat. Transf. 348 (2026).
DOI: 10.1016/j.jqsrt.2025.109687
16. D. H. Marsico, S. Du, S. N. Stechmann. Can second-order numerical accuracy be achieved for moist atmospheric dynamics with non-smoothness at cloud edge? J. Adv. Model. Earth Syst. 17 (2025).
DOI: 10.1029/2025MS005293
15. S. Du and S. N. Stechmann. Element learning: a systematic approach of accelerating finite element-type methods via machine learning, with applications to radiative transfer. J. Comput. Math. (2024).
DOI: 10.4208/jcm.2407-m2024-0047
14. S. Du and S. N. Stechmann. Inverse radiative transfer with goal-oriented hp-adaptive mesh refinement: adaptive-mesh inversion. Inverse Probl. 39 (2023), no. 11.
DOI: 10.1088/1361-6420/acf785
13. B. Cockburn, S. Du, and M. A. Sánchez. A priori error analysis of new semidiscrete, Hamiltonian HDG methods for the time-dependent Maxwell's equations. ESAIM: M2AN 57 (2023), no. 4, 2097–2129.
DOI: 10.1051/m2an/2023048
12. S. Du and S. N. Stechmann. Fast, low-memory numerical methods for radiative transfer via hp-adaptive mesh refinement. J. Comput. Phys. 480 (2023).
DOI: 10.1016/j.jcp.2023.112021
11. S. Du and S. N. Stechmann. A universal predictor-corrector approach for minimizing artifacts due to mesh refinement. J. Adv. Model. Earth Syst. 15 (2023).
DOI: 10.1029/2023MS003688
10. B. Cockburn, S. Du, and M. A. Sánchez. Combining finite element space-discretization with symplectic time-marching schemes for linear Hamiltonian systems. Front. Appl. Math. Stat. 9 (2023).
DOI: 10.3389/fams.2023.1165371
9. M. A. Sánchez, S. Du, B. Cockburn, N.-C. Nguyen, and J. Peraire. Symplectic Hamiltonian finite element methods for electromagnetics. Comput. Methods Appl. Mech. Engrg. 396 (2022).
DOI: 10.1016/j.cma.2022.114969
8. B. Cockburn, M. A. Sánchez, and S. Du. Discontinuous Galerkin methods with time-operators in their numerical traces for time-dependent electromagnetics. Comput. Meth. Appl. Math. 22 (2022), no. 4, 775–796.
DOI: 10.1515/cmam-2021-0215
7. S. Du and F.-J. Sayas. A note on devising HDG+ projections on polyhedral elements. Math. Comp. 90 (2021), 65–79.
DOI: 10.1090/mcom/3573
6. S. Du. HDG methods for the Stokes equation based on strong symmetric stress formulations. J. Sci. Comput. 85, 8 (2020).
DOI: 10.1007/s10915-020-01309-7
5. S. Du and F.-J. Sayas. A unified error analysis of hybridizable discontinuous Galerkin methods for the static Maxwell equations. SIAM J. Numer. Anal. 58 (2020), no. 2, 1367–1391.
DOI: 10.1137/19M1290966

4. S. Du and F.-J. Sayas. New analytical tools for HDG in elasticity, with applications to elastodynamics. Math. Comp. 89 (2020), 1745–1782.
   DOI: 10.1090/mcom/3499
3. S. Du and N. Du. A factorization of least-squares projection schemes for ill-posed problems. Comput. Meth. Appl. Math. 20 (2020), no. 4, 783–798.
   DOI: 10.1515/cmam-2019-0173
2. T.S. Brown, S. Du, H. Eruslu, and F.-J. Sayas. Analysis of models for viscoelastic wave propagation. Appl. Math. Nonlin. Sci. 3 (2018), no. 1, 55–96.
   DOI: 10.21042/AMNS.2018.1.00006

Books

1. S. Du and F.-J. Sayas. An invitation to the theory of the Hybridizable Discontinuous Galerkin Method. SpringerBriefs in Mathematics (2019).
   DOI: 10.1007/978-3-030-27230-2

GRANTS

- NSF (DMS-2608769): Efficient High-Order Methods for Radiative Transfer via Substructuring. PI. $112,085 at Syracuse University (2026–2029).
- NSF (AGS–2326631): Convective Processes in the Tropics Across Scales. Co-PI. $471,155 at UW-Madison (2024–2027).

PRESENTATIONS

Invited talks

36. Accelerating finite element methods via machine learning on a single element
    Mathematical Sciences Colloquium, Rensselaer Polytechnic Institute — Mar 2026
35. Accelerating finite element via machine learning on one element
    Keynote, Winter Research Symposium, University of Delaware — Feb 2026
34. Inverse radiative transfer via goal-oriented adaptive mesh refinement
    SIAM New York–New Jersey–Pennsylvania Section — Nov 2025
33. Accelerating finite element-type methods with machine learning on substructures
    Third HKSIAM Biennial Conference, Chinese University of Hong Kong — Jul 2025
32. Element learning: accelerating finite element-type methods via machine learning, with applications to radiative transfer
    SIAM sectional meeting, Fort Worth, Texas — Mar 2025
31. Forward and inverse computation for radiative transfer via hp-adaptive mesh refinements
    14th AIMS Conference, Abu Dhabi — Dec 2024
30. Inverse radiative transfer via goal-oriented hp-adaptive mesh refinement
    14th AIMS Conference, Abu Dhabi — Dec 2024
29. Element learning: accelerating finite element-type methods via operator learning with cost-effective training
    SIAM sectional meeting, Rochester Institute of Technology — Nov 2024
28. Forward and inverse computation for radiative transfer via hp-adaptive mesh refinement and machine learning acceleration
    Analysis seminar, Binghamton University — Oct 2024

27. Element learning: a systematic approach of accelerating finite element-type methods via machine learning
    CCAM seminar, Purdue University Mar 2024
26. Element learning: a systematic approach of accelerating finite element-type methods via machine learning
    Analysis and Data Science Seminar, SUNY at Albany Feb 2024
25. Element learning: a systematic approach of accelerating finite element-type methods via machine learning
    Math Department Colloquium, Syracuse University Jan 2024
24. Element learning: a systematic approach of accelerating finite element-type methods via machine learning
    Math Department Colloquium, Chinese University of Hong Kong Dec 2023
23. Element learning: a systematic approach of accelerating finite element-type methods, with applications to radiative transfer
    University of Electronic Science and Technology of China Nov 2023
22. Element learning: a systematic approach of accelerating finite element-type methods via machine learning, with applications to radiative transfer
    Scientific Computing Seminars, University of Houston Nov 2023
21. Element learning: a systematic approach of accelerating finite element-type methods via machine learning, with applications to radiative transfer
    Applied Math seminar, University of Louisiana at Lafayette Oct 2023
20. Element learning: a systematic approach of accelerating finite element-type methods, with applications to radiative transfer
    Numerical analysis and PDE seminar, University of Delaware Sep 2023
19. Energy-conserving discontinuous Galerkin methods with time-operators in their traces for time-dependent electromagnetics
    17th USNCCM, Albuquerque, NM Jul 2023
18. Fast, low-memory methods for radiative transfer through hp-adaptive mesh refinement
    13th AIMS Conference, Wilmington, NC Jun 2023
17. Unified analysis of HDG methods for the static Maxwell equations
    CILAMCE-PANACM 2021, Brazil Nov 2021
16. Generalized projection-based error analysis of hybridizable discontinuous Galerkin (HDG) methods
    CEDYA2021, Spain Jun 2021
15. Unified analysis of HDG methods for the static Maxwell equations
    SIAM CSE2021, Virtual Meeting Mar 2021
14. Projection-based analysis of hybridizable discontinuous Galerkin (HDG) methods
    Wenbo Li Prize Talk, U of Delaware Feb 2020
13. New analysis techniques of HDG+ method
    SIAM Sectional Meeting, Iowa State U Oct 2019
12. Uniform-in-time optimal convergent HDG method for transient elastic waves with strong symmetric stress formulation
    WAVES2019, TU Wien, Vienna Aug 2019
11. Hybridizable Discontinuous Galerkin schemes for elastic waves
    ICIAM2019, Valencia Jul 2019
10. HDG for transient elastic waves
    WONAPDE2019, U of Concepción Jan 2019

Contributed talks

9. Element learning: accelerating finite element methods via operator learning
   FEM Circus, U of Notre Dame — Oct 2023
8. Three-dimensional radiative transfer: fast, low-memory numerical methods
   Collective Madison Meeting, Madison, WI — Aug 2022
7. Projection-based analysis of HDG methods with reduced stabilization
   DelMar Num Day 2019, U of Maryland — May 2019
6. Projection-based error analysis of HDG methods for transient elastic waves
   FEM Circus, U of Delaware — Nov 2018
5. Devising a tailored projection for a new HDG method in linear elasticity
   FEM Circus, U of Tennessee — Mar 2018
4. A new HDG projection and its applications
   Mid-Atlantic Numerical Analysis Day, Temple U — Nov 2017

Poster presentation

3. Fast, low-memory numerical methods for radiative transfer: forward and inverse problems
   New Trends in Computational and Data Sciences, Caltech — Dec 2022
2. Hybridizable Discontinuous Galerkin methods in transient elastodynamics
   FACM2018, New Jersey Institute of Technology — Aug 2018
1. Building a computational code for 3D viscoelastic wave simulation
   Mid-Atlantic Numerical Analysis Day, Temple U — Nov 2016

TEACHING

Instructor

- Methods of Numerical Analysis II (MAT684) at Syracuse University — Spring 2026
- Applied Linear Algebra (MAT532) at Syracuse University — Fall 2025
- Multivariable Calculus (MAT397) at Syracuse University — Spring 2025
- Linear Algebra and Differential Equations (Math320) at UW-Madison — Spring 2023

Teaching Assistant

- Analytic Geometry and Calculus C (Math243) — Fall 2016 & Fall 2017
- Analytic Geometry and Calculus B (Math242) — Spring 2017
- Calculus I (Math221) — Spring 2018
- Review of Advanced Mathematical Problems
  (summer courses offered to incoming graduate students) — Fall 2018

MENTORING ACTIVITIES

Graduate mentorship

- Jason Torchinsky (co-mentored with S. N. Stechmann) — 2022 – 2023

Undergraduate mentorship

- SOURCE Bridge Award mentorship, Syracuse University — 2026
- WISCERS project at the University of Wisconsin-Madison — 2023
- GEMS summer research project at the University of Delaware — Fall 2016

JOURNAL REFEREE

Journal of Scientific Computing • SIAM Journal on Scientific Computing • SIAM Multiscale Modeling and Simulation • ESAIM: Mathematical Modelling and Numerical Analysis • Inverse Problems and Imaging • Computers and Mathematics with Applications • Physica D: Nonlinear Phenomena • Frontiers in Applied Mathematics and Statistics

AWARDS AND HONORS

Wenbo Li Prize	2020
University Doctoral Fellowship Award at the University of Delaware	2019
ICIAM2019 travel grant	2019
Graduate Enrichment Fellowship at the University of Delaware	2018
GEMS project fund at the University of Delaware	Summer 2016
National Scholarship for Graduate Students of China	2013
People's Scholarship of Wuhan University	2011
Outstanding Student of Wuhan University	2009 – 2011

SERVICES

- Organizing minisymposium "Recent Developments on Hybrid Methods Combining Neural Networks with Classical Numerical Methods"
  SIAM Pacific Northwest Section — Oct 2025
- Mentor for SIAM Virtual Resume Building Workshop — Oct 2024